

Algebra 1  
Unit 13  
Lesson 11 Practice

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

The table shows the number of tons of blueberries that were imported to the US from 2005 to 2013.

| Year | Blueberries Imported (tons) |
|------|-----------------------------|
| 2005 | 26,319                      |
| 2006 | 32,602                      |
| 2007 | 35,125                      |
| 2008 | 52,111                      |
| 2009 | 60,450                      |
| 2010 | 76,743                      |
| 2011 | 87,336                      |
| 2012 | 96,367                      |
| 2013 | 103,502                     |

1. Determine the average rate of change from 2010 to 2013.
2. Interpret the average rate of change for the interval 2010 to 2013 in terms of the context.
3. Determine the average rate of change from 2005 to 2009.
4. Interpret the average rate of change for the interval from 2005 to 2009 in terms of the context.
5. How does the average rate of change for the interval from 2005 to 2009 compare to the average rate of change for the interval 2010 to 2013?

The function  $M(x) = 5x^2 - 160x + 6179$  models the number of miles, in millions, U.S. passengers traveled on a train for the period of 1970 to 2000, where  $x$  is the number of years since 1970.

6. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1980 to 1985?
7. Interpret the average rate of change for the interval 1980 to 1985 in terms of the context.
8. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1985 to 1990?
9. Interpret the average rate of change for the interval 1985 to 1990 in terms of the context.
10. The graph of a function is shown. Points  $(2, 2.25)$  and  $(5, 7.594)$  are on the function. Calculate the average rate of change for the line that passes through the points  $(2, 2.25)$  and  $(5, 7.594)$ .

